

Drive already mounted at /content/drive; to attempt to forcibly remount, call drive.mount("/content/drive", force_remount=True).

Seeds used:
[42, 52, 62, 72, 82]

Dataset paths:
Train: /content/drive/MyDrive/Bone_Fracture_Binary_Classification/train
Validation: /content/drive/MyDrive/Bone_Fracture_Binary_Classification/val
Test: /content/drive/MyDrive/Bone_Fracture_Binary_Classification/test

Dataset directories found successfully.

Class information:
Train classes: ['fractured', 'not fractured']
Validation classes: ['fractured', 'not fractured']
Test classes: ['fractured', 'not fractured']

Class-to-index mapping:
{'fractured': 0, 'not fractured': 1}

Dataset sizes:
Training: 9246
Validation: 829
Testing: 506

Using Device: cuda

MODEL PARAMETER INFORMATION

Total parameters: 4,012,677
Trainable parameters: 4,012,677
Classical head parameters: 5,129

STARTING SEED 42

```
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted to RGBA mode
  warnings.warn(
Seed 42 | Epoch 1/10 | Train Loss: 0.4031 | Val Loss: 0.3150 | Train Acc: 90.11% | Val Acc: 94.93%
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted to RGBA mode
  warnings.warn(
Seed 42 | Epoch 2/10 | Train Loss: 0.2526 | Val Loss: 0.2710 | Train Acc: 98.90% | Val Acc: 97.35%
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted to RGBA mode
  warnings.warn(
Seed 42 | Epoch 3/10 | Train Loss: 0.2304 | Val Loss: 0.2464 | Train Acc: 99.44% | Val Acc: 98.19%
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted to RGBA mode
  warnings.warn(
Seed 42 | Epoch 4/10 | Train Loss: 0.2193 | Val Loss: 0.2454 | Train Acc: 99.60% | Val Acc: 98.43%
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted to RGBA mode
  warnings.warn(
Seed 42 | Epoch 5/10 | Train Loss: 0.2176 | Val Loss: 0.2399 | Train Acc: 99.69% | Val Acc: 98.55%
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted to RGBA mode
  warnings.warn(
Seed 42 | Epoch 6/10 | Train Loss: 0.2172 | Val Loss: 0.2391 | Train Acc: 99.63% | Val Acc: 98.55%
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted to RGBA mode
  warnings.warn(
Seed 42 | Epoch 7/10 | Train Loss: 0.2161 | Val Loss: 0.2399 | Train Acc: 99.69% | Val Acc: 98.55%
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted to RGBA mode
  warnings.warn(
Seed 42 | Epoch 8/10 | Train Loss: 0.2157 | Val Loss: 0.2388 | Train Acc: 99.69% | Val Acc: 98.67%
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted to RGBA mode
  warnings.warn(
Seed 42 | Epoch 9/10 | Train Loss: 0.2146 | Val Loss: 0.2378 | Train Acc: 99.78% | Val Acc: 98.67%
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted to RGBA mode
  warnings.warn(
Seed 42 | Epoch 10/10 | Train Loss: 0.2148 | Val Loss: 0.2379 | Train Acc: 99.69% | Val Acc: 98.79%
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted to RGBA mode
  warnings.warn(
-----
```

Seed 42 completed.
Time taken for Seed 42: 16.38 minutes
Best Validation Loss: 0.2378

STARTING SEED 52

```
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted to RGBA mode
  warnings.warn(
Seed 52 | Epoch 1/10 | Train Loss: 0.4879 | Val Loss: 0.3860 | Train Acc: 86.30% | Val Acc: 95.05%
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted to RGBA mode
  warnings.warn(
Seed 52 | Epoch 2/10 | Train Loss: 0.3637 | Val Loss: 0.3536 | Train Acc: 98.90% | Val Acc: 96.26%
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted to RGBA mode
  warnings.warn(
Seed 52 | Epoch 3/10 | Train Loss: 0.3373 | Val Loss: 0.3298 | Train Acc: 99.42% | Val Acc: 97.47%
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted to RGBA mode
  warnings.warn(
-----
```

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warnings.warn(
Seed 52 | Epoch 4/10 | Train Loss: 0.3273 | Val Loss: 0.3254 | Train Acc: 99.33% | Val Acc: 97.83%
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted to RGBA mode
warnings.warn(
Seed 52 | Epoch 5/10 | Train Loss: 0.3237 | Val Loss: 0.3235 | Train Acc: 99.52% | Val Acc: 97.71%
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted to RGBA mode
warnings.warn(
Seed 52 | Epoch 6/10 | Train Loss: 0.3214 | Val Loss: 0.3190 | Train Acc: 99.65% | Val Acc: 98.07%
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted to RGBA mode
warnings.warn(
Seed 52 | Epoch 7/10 | Train Loss: 0.3207 | Val Loss: 0.3193 | Train Acc: 99.59% | Val Acc: 97.83%
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted to RGBA mode
warnings.warn(
Seed 52 | Epoch 8/10 | Train Loss: 0.3205 | Val Loss: 0.3182 | Train Acc: 99.63% | Val Acc: 98.07%
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted to RGBA mode
warnings.warn(
Seed 52 | Epoch 9/10 | Train Loss: 0.3208 | Val Loss: 0.3184 | Train Acc: 99.58% | Val Acc: 97.83%
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted to RGBA mode
warnings.warn(
Seed 52 | Epoch 10/10 | Train Loss: 0.3197 | Val Loss: 0.3170 | Train Acc: 99.68% | Val Acc: 97.95%
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted to RGBA mode
warnings.warn(
-----
Seed 52 completed.
Time taken for Seed 52: 8.88 minutes
Best Validation Loss: 0.3170
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```

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STARTING SEED 62

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```
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted to RGBA mode
warnings.warn(
Seed 62 | Epoch 1/10 | Train Loss: 0.4668 | Val Loss: 0.3917 | Train Acc: 88.38% | Val Acc: 94.81%
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted to RGBA mode
warnings.warn(
Seed 62 | Epoch 2/10 | Train Loss: 0.3390 | Val Loss: 0.3550 | Train Acc: 98.68% | Val Acc: 96.38%
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted to RGBA mode
warnings.warn(
Seed 62 | Epoch 3/10 | Train Loss: 0.3132 | Val Loss: 0.3338 | Train Acc: 99.32% | Val Acc: 97.10%
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted to RGBA mode
warnings.warn(
Seed 62 | Epoch 4/10 | Train Loss: 0.3020 | Val Loss: 0.3290 | Train Acc: 99.49% | Val Acc: 97.35%
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted to RGBA mode
warnings.warn(
Seed 62 | Epoch 5/10 | Train Loss: 0.3004 | Val Loss: 0.3276 | Train Acc: 99.53% | Val Acc: 97.35%
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted to RGBA mode
warnings.warn(
Seed 62 | Epoch 6/10 | Train Loss: 0.2976 | Val Loss: 0.3235 | Train Acc: 99.65% | Val Acc: 97.95%
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted to RGBA mode
warnings.warn(
Seed 62 | Epoch 7/10 | Train Loss: 0.2962 | Val Loss: 0.3243 | Train Acc: 99.70% | Val Acc: 97.59%
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted to RGBA mode
warnings.warn(
Seed 62 | Epoch 8/10 | Train Loss: 0.2961 | Val Loss: 0.3249 | Train Acc: 99.69% | Val Acc: 97.23%
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted to RGBA mode
warnings.warn(
Seed 62 | Epoch 9/10 | Train Loss: 0.2962 | Val Loss: 0.3209 | Train Acc: 99.64% | Val Acc: 98.19%
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted to RGBA mode
warnings.warn(
Seed 62 | Epoch 10/10 | Train Loss: 0.2955 | Val Loss: 0.3244 | Train Acc: 99.71% | Val Acc: 97.23%
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted to RGBA mode
warnings.warn(
-----
Seed 62 completed.
Time taken for Seed 62: 8.95 minutes
Best Validation Loss: 0.3209
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```

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STARTING SEED 72

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/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted to RGBA mode
warnings.warn(
Seed 72 | Epoch 1/10 | Train Loss: 0.4225 | Val Loss: 0.3242 | Train Acc: 88.47% | Val Acc: 94.69%
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted to RGBA mode
warnings.warn(
Seed 72 | Epoch 2/10 | Train Loss: 0.2728 | Val Loss: 0.2840 | Train Acc: 98.89% | Val Acc: 96.86%
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted to RGBA mode
warnings.warn(
Seed 72 | Epoch 3/10 | Train Loss: 0.2516 | Val Loss: 0.2590 | Train Acc: 99.29% | Val Acc: 97.83%
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted to RGBA mode
warnings.warn(
Seed 72 | Epoch 4/10 | Train Loss: 0.2398 | Val Loss: 0.2565 | Train Acc: 99.60% | Val Acc: 98.19%
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted to RGBA mode
warnings.warn(
Seed 72 | Epoch 5/10 | Train Loss: 0.2393 | Val Loss: 0.2536 | Train Acc: 99.50% | Val Acc: 98.19%
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted to RGBA mode
warnings.warn(
Seed 72 | Epoch 6/10 | Train Loss: 0.2367 | Val Loss: 0.2515 | Train Acc: 99.62% | Val Acc: 98.19%
```

```

/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted to RGBA mode
  warnings.warn(
Seed 72 | Epoch 8/10 | Train Loss: 0.2346 | Val Loss: 0.2511 | Train Acc: 99.72% | Val Acc: 98.31%
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted to RGBA mode
  warnings.warn(
Seed 72 | Epoch 8/10 | Train Loss: 0.2346 | Val Loss: 0.2508 | Train Acc: 99.72% | Val Acc: 98.31%
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted to RGBA mode
  warnings.warn(
Seed 72 | Epoch 9/10 | Train Loss: 0.2356 | Val Loss: 0.2508 | Train Acc: 99.62% | Val Acc: 98.19%
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted to RGBA mode
  warnings.warn(
Seed 72 | Epoch 10/10 | Train Loss: 0.2351 | Val Loss: 0.2506 | Train Acc: 99.64% | Val Acc: 98.31%
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted to RGBA mode
  warnings.warn(

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Seed 72 completed.
Time taken for Seed 72: 9.06 minutes
Best Validation Loss: 0.2506
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```

STARTING SEED 82

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/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted to RGBA mode
  warnings.warn(
Seed 82 | Epoch 1/10 | Train Loss: 0.4463 | Val Loss: 0.3416 | Train Acc: 88.20% | Val Acc: 95.05%
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted to RGBA mode
  warnings.warn(
Seed 82 | Epoch 2/10 | Train Loss: 0.2975 | Val Loss: 0.3059 | Train Acc: 98.84% | Val Acc: 96.26%
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted to RGBA mode
  warnings.warn(
Seed 82 | Epoch 3/10 | Train Loss: 0.2751 | Val Loss: 0.2780 | Train Acc: 99.30% | Val Acc: 98.31%
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted to RGBA mode
  warnings.warn(
Seed 82 | Epoch 4/10 | Train Loss: 0.2630 | Val Loss: 0.2748 | Train Acc: 99.61% | Val Acc: 98.43%
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted to RGBA mode
  warnings.warn(
Seed 82 | Epoch 5/10 | Train Loss: 0.2631 | Val Loss: 0.2704 | Train Acc: 99.43% | Val Acc: 98.55%
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted to RGBA mode
  warnings.warn(
Seed 82 | Epoch 6/10 | Train Loss: 0.2596 | Val Loss: 0.2664 | Train Acc: 99.57% | Val Acc: 98.79%
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted to RGBA mode
  warnings.warn(
Seed 82 | Epoch 7/10 | Train Loss: 0.2580 | Val Loss: 0.2657 | Train Acc: 99.69% | Val Acc: 98.79%
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted to RGBA mode
  warnings.warn(
Seed 82 | Epoch 8/10 | Train Loss: 0.2578 | Val Loss: 0.2661 | Train Acc: 99.70% | Val Acc: 98.91%
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted to RGBA mode
  warnings.warn(
Seed 82 | Epoch 9/10 | Train Loss: 0.2571 | Val Loss: 0.2655 | Train Acc: 99.76% | Val Acc: 98.79%
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted to RGBA mode
  warnings.warn(
Seed 82 | Epoch 10/10 | Train Loss: 0.2577 | Val Loss: 0.2654 | Train Acc: 99.68% | Val Acc: 98.79%
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted to RGBA mode
  warnings.warn(

```

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Seed 82 completed.
Time taken for Seed 82: 9.09 minutes
Best Validation Loss: 0.2654
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ALL FIVE SEEDS COMPLETED

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Total experiment time: 52.36 minutes
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PER-SEED PERFORMANCE

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Seed   Dataset      Metric      Value
42     Training      Accuracy (%) 99.859399
42     Validation    Accuracy (%) 98.673100
42     Testing        Accuracy (%) 98.616601
42     Training      Sensitivity (%) 99.935345
42     Validation    Sensitivity (%) 99.390244
42     Testing        Sensitivity (%) 98.880597
42     Training      Specificity (%) 99.782892
42     Validation    Specificity (%) 97.626113
42     Testing        Specificity (%) 98.319328
42     Training      Precision (%) 99.784807
42     Validation    Precision (%) 98.390342
42     Testing        Precision (%) 98.513011
42     Training      F1-Score (%) 99.860019
42     Validation    F1-Score (%) 98.887765
42     Testing        F1-Score (%) 98.696462
42     Training      Error Rate (%) 0.140601
42     Validation    Error Rate (%) 1.326900
42     Testing        Error Rate (%) 1.383399
42     Training      ROC-AUC      0.999963

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42	Validation	ROC-AUC	0.998329
42	Testing	ROC-AUC	0.998981
52	Training	Accuracy (%)	99.794506
52	Validation	Accuracy (%)	97.949337
52	Testing	Accuracy (%)	98.023715
52	Training	Sensitivity (%)	99.784483
52	Validation	Sensitivity (%)	98.780488
52	Testing	Sensitivity (%)	97.761194
52	Training	Specificity (%)	99.804603
52	Validation	Specificity (%)	96.735905
52	Testing	Specificity (%)	98.319328
52	Training	Precision (%)	99.805993
52	Validation	Precision (%)	97.786720
52	Testing	Precision (%)	98.496241
52	Training	F1-Score (%)	99.795237
52	Validation	F1-Score (%)	98.281092
52	Testing	F1-Score (%)	98.127341
52	Training	Error Rate (%)	0.205494
52	Validation	Error Rate (%)	2.050663
52	Testing	Error Rate (%)	1.976285
52	Training	ROC-AUC	0.999625
52	Validation	ROC-AUC	0.995688
52	Testing	ROC-AUC	0.992122
62	Training	Accuracy (%)	99.870214
62	Validation	Accuracy (%)	98.190591
62	Testing	Accuracy (%)	98.023715
62	Training	Sensitivity (%)	99.870690
62	Validation	Sensitivity (%)	98.780488
62	Testing	Sensitivity (%)	97.761194
62	Training	Specificity (%)	99.869735
62	Validation	Specificity (%)	97.329377
62	Testing	Specificity (%)	98.319328
62	Training	Precision (%)	99.870690
62	Validation	Precision (%)	98.181818
62	Testing	Precision (%)	98.496241
62	Training	F1-Score (%)	99.870690
62	Validation	F1-Score (%)	98.480243
62	Testing	F1-Score (%)	98.127341
62	Training	Error Rate (%)	0.129786
62	Validation	Error Rate (%)	1.809409
62	Testing	Error Rate (%)	1.976285
62	Training	ROC-AUC	0.999797
62	Validation	ROC-AUC	0.995881
62	Testing	ROC-AUC	0.993541
72	Training	Accuracy (%)	99.848583
72	Validation	Accuracy (%)	98.311218
72	Testing	Accuracy (%)	97.233202
72	Training	Sensitivity (%)	99.806034
72	Validation	Sensitivity (%)	97.967480
72	Testing	Sensitivity (%)	96.268657
72	Training	Specificity (%)	99.891446
72	Validation	Specificity (%)	98.813056
72	Testing	Specificity (%)	98.319328
72	Training	Precision (%)	99.892148
72	Validation	Precision (%)	99.176955
72	Testing	Precision (%)	98.473282
72	Training	F1-Score (%)	99.849073
72	Validation	F1-Score (%)	98.568507
72	Testing	F1-Score (%)	97.358491
72	Training	Error Rate (%)	0.151417
72	Validation	Error Rate (%)	1.688782
72	Testing	Error Rate (%)	2.766798
72	Training	ROC-AUC	0.999769
72	Validation	ROC-AUC	0.995893
72	Testing	ROC-AUC	0.992804
82	Training	Accuracy (%)	99.848583
82	Validation	Accuracy (%)	98.793727
82	Testing	Accuracy (%)	98.221344
82	Training	Sensitivity (%)	99.870690
82	Validation	Sensitivity (%)	98.983740
82	Testing	Sensitivity (%)	98.134328
82	Training	Specificity (%)	99.826314
82	Validation	Specificity (%)	98.516320
82	Testing	Specificity (%)	98.319328
82	Training	Precision (%)	99.827660
82	Validation	Precision (%)	98.983740
82	Testing	Precision (%)	98.501873
82	Training	F1-Score (%)	99.849170
82	Validation	F1-Score (%)	98.983740
82	Testing	F1-Score (%)	98.317757
82	Training	Error Rate (%)	0.151417
82	Validation	Error Rate (%)	1.206273
82	Testing	Error Rate (%)	1.778656
82	Training	ROC-AUC	0.999976
82	Validation	ROC-AUC	0.998703
82	Testing	ROC-AUC	0.997727

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PERFORMANCE COMPARISON TABLE - MEAN ± SD
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Metric	Training	Validation	Testing
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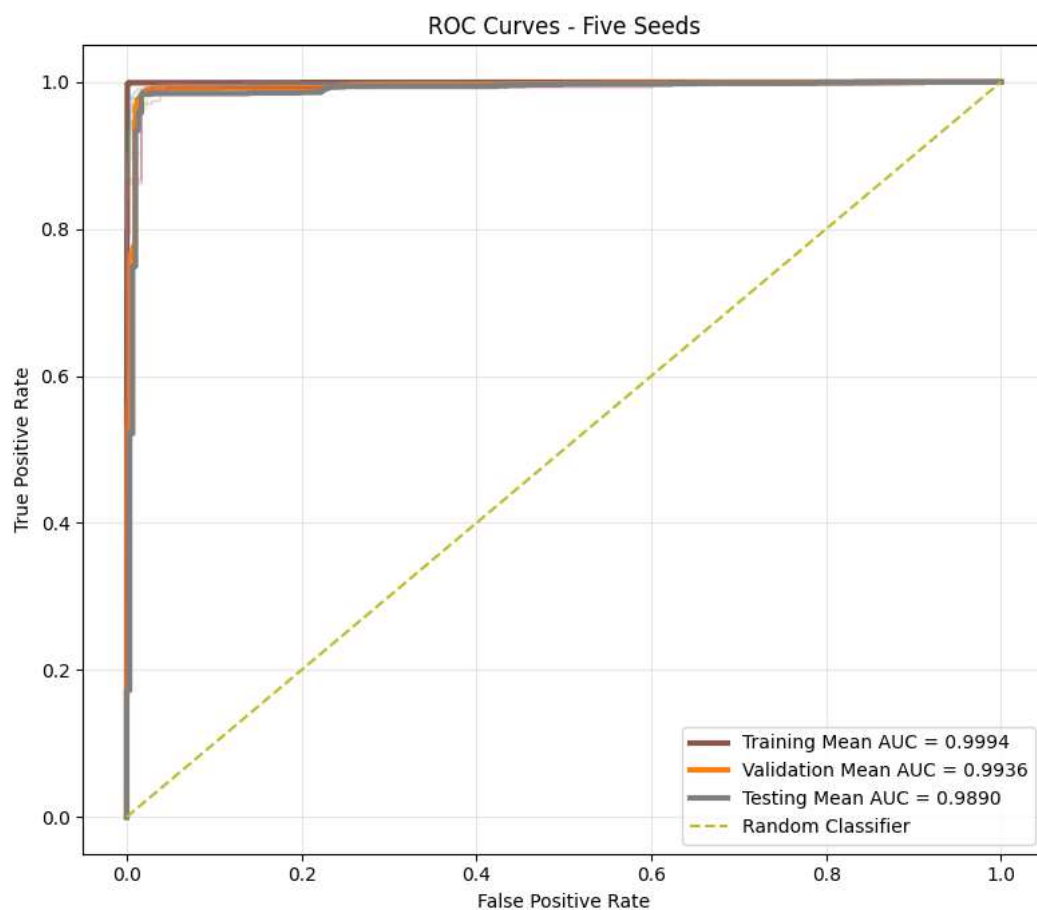
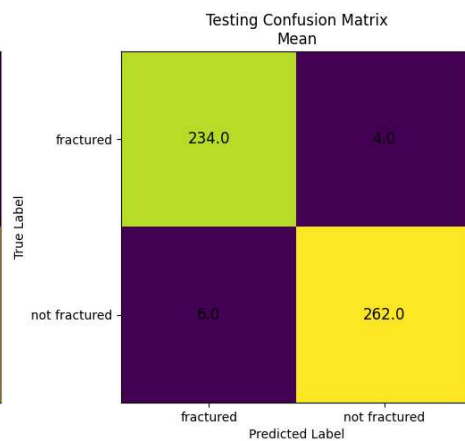
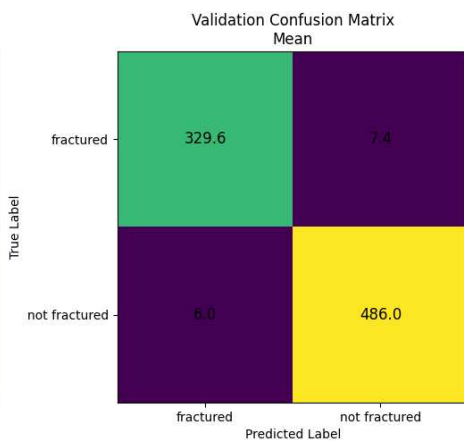
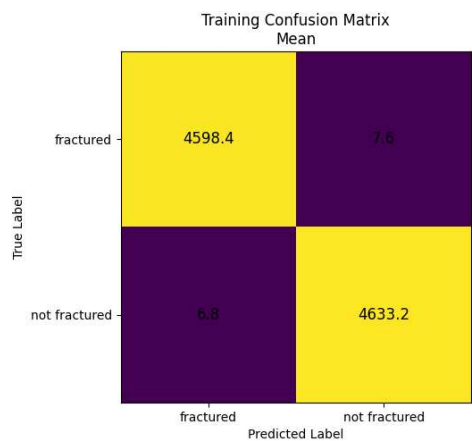
Accuracy (%)	99.84 ± 0.03	98.38 ± 0.35	98.02 ± 0.50
Sensitivity (%)	99.85 ± 0.06	98.78 ± 0.52	97.76 ± 0.95
Specificity (%)	99.83 ± 0.05	97.80 ± 0.85	98.32 ± 0.00
Precision (%)	99.84 ± 0.04	98.50 ± 0.57	98.50 ± 0.01
F1-Score (%)	99.84 ± 0.03	98.64 ± 0.29	98.13 ± 0.49
Error Rate (%)	0.16 ± 0.03	1.62 ± 0.35	1.98 ± 0.50
ROC-AUC	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00

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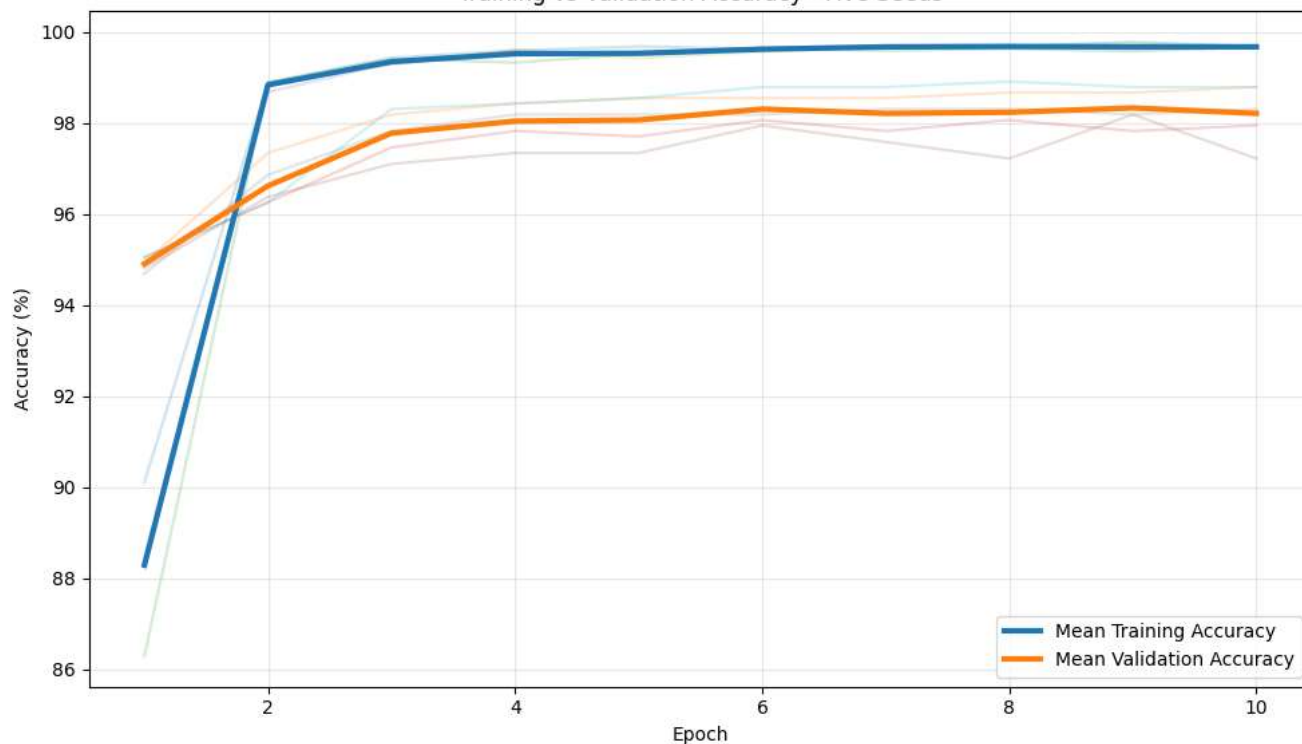
TIME TAKEN FOR EACH SEED

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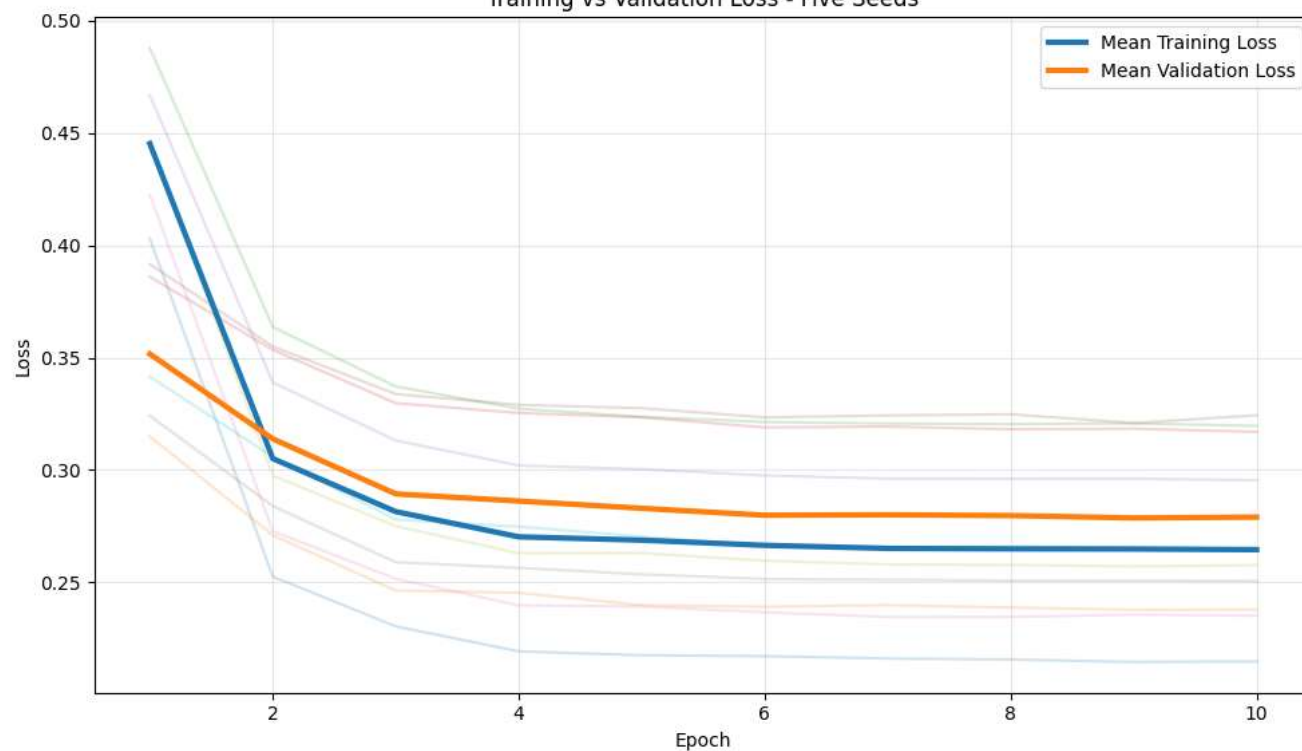
Seed	Time (minutes)
42	16.381140
52	8.884036
62	8.947064
72	9.058674
82	9.090579



Training vs Validation Accuracy - Five Seeds



Training vs Validation Loss - Five Seeds



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FINAL MODEL SUMMARY

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Model:

EfficientNet-B0 → 1280 features → Linear(1280→4) → Tanh → Linear(4→1)

Seeds:

[42, 52, 62, 72, 82]

Total parameters:

4,012,677

Trainable parameters:

4,012,677

Classical head parameters:

5,129

Total experiment time:

52.36 minutes

Performance:

Metric	Training	Validation	Testing
Accuracy (%)	99.84 ± 0.03	98.38 ± 0.35	98.02 ± 0.50
Sensitivity (%)	99.85 ± 0.06	98.78 ± 0.52	97.76 ± 0.95
Specificity (%)	99.83 ± 0.05	97.80 ± 0.85	98.32 ± 0.00
Precision (%)	99.84 ± 0.04	98.50 ± 0.57	98.50 ± 0.01
F1-Score (%)	99.84 ± 0.03	98.64 ± 0.29	98.13 ± 0.49
Error Rate (%)	0.16 ± 0.03	1.62 ± 0.35	1.98 ± 0.50
ROC-AUC	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00

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EXPERIMENT COMPLETED SUCCESSFULLY

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